MYAUDIOSHELF

Full Product Architecture & UI/UX Build Specification

Flask + Jinja audiobook commerce, streaming and production platform with Stripe guest checkout

Primary UI reference: NaijaHub.ai's overall visual language and interaction quality - typography hierarchy, editorial spacing, solid surfaces, restrained motion, navigation proportions and responsive polish - adapted for an audiobook storefront. Do not use numbered homepage sections. Do not reproduce NaijaHub branding or content.

Brand direction: lemon green accent, warm light surfaces, dark typography, premium book imagery, strong editorial composition, no gradients, no glassmorphism, no generic AI aesthetic.

Design System & NaijaHub-Inspired Direction

MyAudioShelf should feel related to the visual sophistication of NaijaHub without becoming a clone. Codex should use NaijaHub as the reference for composition, spacing, typography scale, navigation behavior, responsive refinement and restrained animations, then reinterpret those principles for books, audio and commerce.

Non-negotiable visual rules

- Do not number homepage sections. No '01', '02', '03' labels and no numbered editorial markers.
- Use a full-width top navigation instead of NaijaHub's sidebar.
- Keep the interface predominantly flat and editorial: solid backgrounds, thin dividers, typography, imagery and whitespace.
- No gradients anywhere in the core visual system.
- No glassmorphism, glowing AI elements, floating orbs, futuristic grid backgrounds or unnecessary 3D effects.
- Lemon green (#C7F000) is the primary accent for important CTAs, active states, player progress and selected highlights.
- Use dark charcoal/near-black typography and warm off-white page surfaces.
- Avoid excessive rounded cards. Use subtle corner radii only where they improve usability.
- Book covers remain visually dominant and preserve natural portrait proportions.
- Animations should be subtle: content reveal, image hover scale, navigation transitions, player transitions and lightweight scroll motion.

Typography

- Inspect NaijaHub's current typographic hierarchy and approximate its relationship between display headings, body copy, metadata, labels and buttons.
- Use large confident headings with controlled line-height rather than oversized decorative type everywhere.
- Keep book metadata compact and readable.
- Use consistent uppercase micro-labels only when useful; never turn every heading into a label.
- Maintain excellent mobile type scaling and readable line lengths.

Homepage - Books Must Dominate

The homepage must function first as an audiobook storefront and discovery experience. Marketing copy supports the catalogue; it must not push published books far down the page.

Homepage area	Implementation
Hero	Large editorial hero with a selected featured audiobook, cover art, short positioning statement, autho
Featured Audiobooks	Curated admin-controlled row/grid of prominent published titles.
Published Books	Large database-driven catalogue preview showing approximately 8-12 live titles, with a clear View Al
New Releases	Recently published titles, ordered by release/publish date.
Listen Before You Buy	Strong preview-led area encouraging visitors to play 60-second trailers.
Listen to the Difference	Selected traditional vs immersive audio comparison demonstrating MyAudioShelf production quality.
Children's Shelf	Age-appropriate children's books with cover, age range, narrator, duration, trailer and price.
MyAudioShelf Originals	Original/owned productions curated separately.
Browse by Genre	Fiction, Non-fiction, Drama, Romance, Children's, Educational, Originals and admin-managed genres.
Immersive Productions	Editorial explanation of enhanced/dramatised productions supported by real title examples.
For Authors & Publishers	Production-service proposition and Submit Your Book / Start a Project CTA.
Behind the Sound	Narrators, cast, producers, sound designers, production stories and credits.
Newsletter	Join the Shelf opt-in.

Important: none of these homepage sections should display numeric prefixes or section numbers. Hierarchy comes from typography, spacing, dividers and content composition.

Published Book Presentation

Book cards

- Cover art should be the strongest element.
- Show title, author, narrator/cast where useful, production type, price and compact trailer action.
- Do not overcrowd cards with every metadata field.
- Use hover/focus states inspired by NaijaHub's restrained interaction style.
- Admin determines Featured, New Release, Children's, Original and homepage visibility.
- Only published and saleable books appear in commerce sections.

Book detail page

- Large cover artwork with title, author, narrator/cast, price and production style.
- Prominent 60-second trailer player.
- Buy Now and Add to Cart without requiring sign-in.
- Synopsis, genre, language, duration, release date and age range where applicable.
- Production notes explaining immersive treatment where relevant.
- Narrator/cast and production credits.
- Related books and more from the same author/genre.
- Clear explanation of what the purchaser receives and how listening access works.

Browse page

- Search title, author, narrator/cast and keyword.
- Filter genre, language, production style, age range, duration and price.
- Sort newest, featured, title, price and popularity when genuine data exists.
- Use the NaijaHub-inspired content density and whitespace rather than a generic e-commerce marketplace grid.

Customer Journey & Guest Commerce

A visitor should be able to discover a book, listen to its trailer, add it to cart and complete Stripe payment without creating an account.

- Cart persists in a signed Flask session.
- Digital audiobook quantity is one per title.
- Cart line items include cover, title, author, price, remove action and trailer control.
- Trailer remains playable from the cart.
- Checkout collects/uses customer email and redirects to Stripe Checkout.
- Backend recalculates authoritative prices from the database before creating the Stripe session.
- Backend creates a pending internal order before Stripe handoff.
- Stripe webhook verifies payment and creates purchase entitlements.
- Success-page arrival alone never unlocks full audio.
- Customer receives a secure Listen Now email.
- Returning guest can request a magic link using the purchase email.
- Optional account creation can be offered later and can claim purchases made with the same verified email.

Protected listening

- Full audiobook files must never live as predictable public static URLs.
- Use entitlement checks and signed/expiring media delivery.
- Support chapter files or a full-book file.
- Remember listening progress.
- Playback speed controls, chapter navigation, seek, volume and resume.
- Use a polished sticky/compact player when appropriate without imitating Spotify unnecessarily.

Admin Experience

The admin experience should be practical and fast. Staff must be able to publish a complete audiobook without editing source code.

Module	Capabilities
Dashboard	Revenue, orders, recent purchases, top titles, trailer activity, new production enquiries, drafts and qui
Audiobooks	Create, edit, preview, publish, unpublish and archive.
Book media	Upload cover/thumbnail, dedicated trailer, full audiobook or ordered chapter files.
Metadata	Title, slug, author, narrator/cast, publisher, genre, language, age range, duration, production style, de
Commerce	Price, currency, saleability and optional promotional pricing.
Homepage curation	Featured books, Originals, Children's titles, hero title, section ordering/visibility where configured.
Authors & cast	Profiles, bios, images and linked works.
Categories	Genres and catalogue taxonomy.
Orders	Customer email, purchased items, totals, Stripe references and payment state.
Entitlements	Customer access state and legitimate revoke/reissue tools.
Production enquiries	Review submissions from authors/publishers and manage status/notes.
SEO	SEO title, description, social image, publication metadata and indexability controls.
Analytics	Sales, book views, trailer plays, conversion and listening activity.
Staff	Role-based admin permissions and audit history.

Add-book workflow

- Create draft.
- Enter metadata and synopsis.
- Upload cover.
- Upload dedicated 60-second trailer.
- Upload full audio or chapters.
- Set price and currency.
- Assign genre, production type and relevant collections.
- Preview the public page.
- Validate required assets/fields.
- Publish.

Narrative Audio Map - Future Production Capability

Preserve the client's proposed Narrative Audio Map as a future production-assistance module. It should analyse or structure scenes, environments, characters, narrative actions, emotional context and transitions, then recommend ambience, effects, intensity, positioning, transitions, narration/background balance and continuity.

- Structure recommendations by book, chapter and scene.
- Record environment, characters, action and emotional context.
- Recommend ambience, sound effects, music/cues, intensity and spatial notes.
- Record transitions and narration/background balance.
- Maintain continuity notes across chapters.
- Every recommendation must support Accept, Modify and Reject.
- Approved creative decisions are stored separately from generated/recommended decisions.
- Do not present this future capability on the public website as complete until it actually exists.

Flask + Jinja Architecture

Recommended structure

```
app/__init__.py
app/config.py
app/extensions.py
app/models/
app/blueprints/main/
app/blueprints/catalog/
app/blueprints/cart/
app/blueprints/checkout/
app/blueprints/library/
app/blueprints/authors/
app/blueprints/admin/
app/services/stripe_service.py
app/services/media_service.py
app/services/email_service.py
app/services/access_service.py
app/templates/
app/static/css/
app/static/js/
migrations/
tests/
.env.example
```

- Use Flask application factory and modular Blueprints.
- Use SQLAlchemy + Flask-Migrate.
- Keep Stripe, email, storage and entitlement logic in services rather than route functions.
- Server-render catalogue and editorial pages for SEO and performance.
- Use JavaScript progressively for audio player, cart enhancement and interactive motion.
- Keep the visual system in reusable CSS variables/components rather than page-specific styling.

Core Data Model

Model	Purpose
Audiobook	Metadata, price, currency, production style, publication state, featured/original/children flags.
Author	Author identity, bio, image and linked books.
Category	Genre/taxonomy and ordering.
MediaAsset	Cover, trailer, full audio, chapter audio and storage metadata.
Chapter	Ordered chapter metadata and audio reference.
Order	Guest email, totals, status and Stripe identifiers.
OrderItem	Immutable purchase snapshot of title and price.
PurchaseEntitlement	Defines access to a purchased audiobook.
MagicLink	Secure guest purchase/library access.
ListeningProgress	Position/chapter progress for purchased audio.
ProductionEnquiry	Author/publisher submission and staff workflow.
AdminUser	Staff identity and role.
AuditLog	Important admin actions.
StripeEvent	Webhook idempotency and processing history.

Money should be stored in integer minor units plus currency, not floating-point values.

Security, SEO & Quality

Security

- CSRF on browser state-changing forms; Stripe webhook uses signature verification.
- Secure, HttpOnly and appropriate SameSite cookies; HTTPS-only in production.
- Rate-limit admin login, magic-link requests, forms and checkout creation.
- Validate file type, MIME type and size for uploads.
- Private full-audio storage with authorization before delivery.
- Stripe webhook idempotency.
- Never store raw card data.
- Never expose secrets in templates, JavaScript or logs.
- Role-based admin authorization and audit logging.

SEO & accessibility

- Unique metadata for each book, author and category.
- XML sitemap and robots.txt.
- Open Graph/social metadata using book artwork.
- Structured data where appropriate.
- Semantic headings, keyboard-accessible audio controls and visible focus states.
- Trailer transcripts where supplied.
- Strong contrast: avoid white text directly on bright lemon green when contrast is insufficient.
- Respect prefers-reduced-motion.

Codex Master Instruction

Build MyAudioShelf as a production-minded Flask + Jinja2 audiobook commerce, streaming and production platform using SQLAlchemy, Flask-Migrate and Stripe Checkout. Use https://www.naijahub.ai as the PRIMARY UI/UX reference for overall visual sophistication: typography hierarchy, spacing rhythm, editorial composition, solid-surface treatment, navigation proportions, responsive behavior, restrained animations, hover behavior and content density. Do NOT copy NaijaHub branding, copy, proprietary assets or its sidebar. MyAudioShelf must use a full-width top navigation. IMPORTANT: do NOT number homepage sections. Do not add 01, 02, 03, numbered labels, numbered editorial markers or similar visual numbering. Section hierarchy must come from typography, whitespace, dividers, imagery and layout. Preserve MyAudioShelf's lemon-green brand accent (#C7F000). Use warm off-white/light neutral backgrounds and near-black typography. Do not use gradients, glassmorphism, glowing AI effects, futuristic orbs, generic AI-dashboard styling, unnecessary 3D elements or excessive rounded cards. The homepage must prioritize ACTUAL PUBLISHED AUDIOBOOKS. Do not make it mostly marketing copy. Include a featured audiobook hero, a major Featured Audiobooks area, a large Published Books area showing roughly 8-12 database-driven published titles, New Releases, Listen Before You Buy, Listen to the Difference, Children's Shelf, MyAudioShelf Originals, Browse by Genre, Immersive Productions, For Authors & Publishers, Behind the Sound and newsletter signup. None of these sections should be numbered. Every book card should prioritize cover art, title and essential metadata. Every book detail page must support a dedicated maximum 60-second public trailer, price, Buy Now, Add to Cart, synopsis, author, narrator/cast, duration, genre, language, production style and related titles. Users MUST be able to buy without creating an account. Implement a signed Flask-session cart and Stripe Checkout. Recalculate product prices server-side. Create an internal pending order before Stripe handoff. Verify Stripe webhook signatures and process events idempotently. Only a verified successful payment creates purchase entitlements. Never grant access merely because the browser reaches the success URL. After purchase, email the guest a secure Listen Now link. Implement My Purchases through secure email magic links so returning buyers can access their books without a password. Full audiobook media must be protected through entitlement checks and signed/expiring delivery; do not expose full audio as public static files. Create an operational admin CMS. Staff must be able to create a draft audiobook, enter metadata, upload cover art, upload a dedicated trailer, upload full audio or chapter audio, set price/currency, assign categories/collections, preview, publish/unpublish/archive, curate homepage books, inspect orders and Stripe status, manage customer entitlements and review author/publisher production enquiries. Keep the proposed Narrative Audio Map as a future production module. Architect structured recommendations for scenes, environments, characters, narrative actions, emotional context, ambience, effects, intensity, positioning, transitions, narration/background balance and continuity. Preserve explicit human Accept / Modify / Reject control. Do not market this capability as live until implemented. Use modular Flask Blueprints, service-layer business logic, SQLAlchemy models, migrations, CSRF protection, secure cookies, rate limiting, upload validation, admin authorization, audit logging, SEO metadata, sitemap, accessibility, responsive layouts and automated tests. Animations should be tasteful and restrained: subtle page/section reveals, image hover scale, navigation transitions, player transitions and micro-interactions. Respect prefers-reduced-motion. The finished site should feel premium, editorial, modern and human - not like an AI-generated SaaS template.

Implementation priority

- Foundation and reusable design system.
- Catalogue, book pages and admin publishing.
- Trailer/player behavior.
- Cart and guest Stripe checkout.
- Webhook-driven orders and entitlements.
- Magic-link purchase library and protected listening.
- Author/publisher production enquiries.
- Homepage curation and published-book discovery.
- SEO, accessibility, security, analytics and testing.
- Narrative Audio Map only as a later module unless separately commissioned.